LLM-MovieLens: A Density-Paradigm Framework with Class-Level Triangulation for LLM-Augmented Collaborative Filtering

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Submission Id: XXX

Abstract

LLM-MovieLens is an extensible testbed for LLM-augmented recommendation that establishes a density-paradigm framework: content-augmentation methods organise into three classes—*regularizers* (auxiliary alignment loss), *replacers* (content-driven embedding generation), and *injection* (additive content bias)—plus an orthogonal *sequential* axis. We triangulate every class-level claim with two structurally-different instantiations per content class. Across four per-item-density datapoints spanning a 90× range, including a same-domain density control, the regularizer-versus-injection gap is monotone in inverse density for *both* regularizer instantiations, swinging from a slight loss at dense scale to roughly +26 to +30% at the sparse extreme; replacers never beat injection and collapse when interactions are sparse (−22% at the sparsest density, $p < 0.001$); injection is density-robust; and the sequential baseline shows the steepest dense-to-sparse decay of any paradigm (6.8×). Within the dense MovieLens-20M setting, controlled ablations show LLM-synthesised profiles beat pure collaborative filtering including the contrastive methods, beat raw genome features at both matched and higher dimensionality (ruling out a dimensionality confound), and beat title-only text at the same encoder (isolating reasoning from encoding), all at $p < 0.05$. Adding a mood vector is statistically null on top-10 ranking but, released as a separate controllable-recommendation primitive, enables zero-shot axis-conditioned retrieval at order-of-magnitude better per-dimension efficiency than genome or title baselines. We release 10,381 movie profiles plus parallel Amazon-Books and MovieLens-1M packs, multi-encoder embeddings, 16 model configurations, training and evaluation infrastructure, a 3-annotator human evaluation, and cross-encoder and cross-provider sensitivity studies. All framework claims are falsifiable on future densities and instantiations.

CCS Concepts

• **Information systems** → **Recommender systems**; *Personalization*; Data mining; • **Computing methodologies** → *Natural language generation*.

Keywords

Recommender systems; large language models; content features; collaborative filtering; benchmark; MovieLens

1 Introduction

Collaborative filtering (CF) is the backbone of modern recommendation systems, learning user preferences from interaction histories alone. Graph neural network (GNN) approaches such as LightGCN [9] and its contrastive variants [4, 34, 35] have achieved state-of-the-art performance by propagating collaborative signals through user–item bipartite graphs. However, CF methods suffer from inherent limitations when items have sparse interaction histories—the *cold-start problem* [31]—where content features become essential for producing meaningful representations.

The MovieLens 20M (ML-20M) dataset [8] is a canonical benchmark in this setting, providing both dense interaction data (20,000,263 ratings from 138,493 users) and rich content metadata in the form of *genome tag relevance scores*: a 1,128-d continuous vector per movie capturing crowd-sourced semantic attributes such as `thought-provoking: 0.91` or `dark comedy: 0.87`. These genome vectors are commonly used as item content features via dimensionality reduction (e.g., PCA to 128-dim), but they remain *flat*: each dimension is treated independently, with no mechanism for reasoning about how tags interact or what they collectively imply about a film's identity.

Large language models (LLMs) offer a fundamentally different approach: rather than using metadata as a fixed numeric vector, an LLM can *read* genome tags, plot synopsis, cast, and reviews, then *synthesize* a holistic characterization that captures inter-attribute relationships no linear method can. For example, given tags `dark comedy`, `satire`, `social commentary`, and `violence`, an LLM can reason that their combination indicates a specific subgenre (political satire with dark humor) and produce text that, when encoded by a sentence transformer, yields a more discriminative embedding than any linear combination of the four scores. Despite this promise, to our knowledge, no prior work has applied LLM-based feature engineering to the full ML-20M dataset or used its genome tags as structured LLM input: existing LLM-for-recommendation methods [23, 31, 32] operate on ML-1M, filtered ML-10M subsets, or Amazon data. We attribute this gap to (a) engineering effort to integrate genome tags with LLM prompts at scale and (b) an untested assumption that genome vectors are already good enough—an assumption we directly challenge. The framework's most immediate downstream application is a density-adaptive paradigm-switcher trained on the released testbed.

Why ML-20M in 2026? Despite its 2015 cutoff, ML-20M remains the field's de-facto benchmark on three concrete grounds: (i) *Richest public structured content annotation*—the 1,128-d genome tag vector [15, 16, 27] has no widely-used public tag-genome equivalent on Amazon or Yelp, and 2018–2026 multimodal augmentations explicitly target ML-20M [5, 6, 26, 37]. (ii) *Density middle ground for graph-CF evaluation*—at 0.91% density (vs. Amazon-Books 0.004%, Yelp 0.013%), ML-20M is the canonical regime where the LightGCN family and its recent contrastive variants are validated. (iii) *Cross-paper comparability + protocol stability*—the post-2020 full-ranking protocol [17] we adopt (§3.5) inherits hundreds of comparable prior results.

Contributions. LLM-MovieLens is an extensible benchmark for LLM-augmented recommendation research. We organize the contributions by what the released experiments scientifically establish:

(1) **A density-paradigm framework for LLM-for-RecSys**, synthesised from four density datapoints with two-instantiation triangulation of both content paradigm classes and presented as a falsifiable cross-density hypothesis. Per-item density determines the winner: *regularizers* R1-gene gain monotonically with sparsity (4-point chain $-1.1\%/+3.7\%/+17.0\%/+29.5\%$ vs. M7), independently corroborated by R1-plus across the full 4-datapoint chain ($-5.6\%/+10.9\%/+16.3\%/+26.4\%$ vs. M7—both regularizer instantiations monotone in 1/density, including the same-domain density control); *replacers* (R2 + R3) never beat M7 and collapse at sparse density, with R2−R3 sign reversal across density extremes; *injection* (M4/M7) is density-robust; *sequential* SASRec is an orthogonal axis with 6.8× dense-to-sparse drop.

(2) **Released artifact (testbed).** 10,381 LLM profiles for ML-20M (text + 10-axis mood + 3–5 themes), parallel profiles for Amazon-Books (9,289) and an ML-1M subset (2,788); pre-computed embeddings under multiple sentence transformers; 16 model configs plus cross-density baselines; training/eval infrastructure (temporal splits, full-ranking, 5-seed, cold-start stratification, resume-capable); 500-profile 3-annotator human evaluation. A new encoder, LLM, or RS method plugs in via a single path override.

(3) **Within-ML-20M ablations and a controllable-recommendation primitive.** LLM profiles beat genome at PCA-128d (+2.5%, $p = 0.001$) and raw 1128-d (+2.9%, $p = 0.022$, ruling out dimensionality), and BERT-on-title at the same encoder (+3.3%, $p = 0.002$); rankings preserved under GPT-4o-mini retrain (residual -0.73%, $p = 0.026$). The 10-axis named mood vector enables tasks beyond top-K NDCG: 30.8% zero-shot controllability, 8.2 × /92× per-dim compactness vs. genome-PCA/BERT-on-title, four cross-LLM-stable axes ($r \geq 0.7$).

Anonymous review artifacts. Code: https://github.com/anonyauthor4review-png/llm-movielens. Data: https://huggingface.co/datasets/anonyauthor4review/llm-movielens. All headline numbers in Tables 4–6 regenerate in one command from the released checkpoints via `make verify-headline` (53-cell verifier). URLs flip to public, non-anonymous versions at camera-ready.

2 Related Work

Content features in CF. Genome tags [27] have served as side features in neural CF [3, 10, 24]; recent graph-contrastive methods (SimGCL [35], XSimGCL [34], LightGCL [4]) push pure-CF without content reasoning.

LLMs for recommendation. *LLM-as-ranker* [11] faces prohibitive serving cost. *LLM-as-feature-engineer* methods include LLMRec [31] (description augmentation + synthetic users), RLMRec [23] (CF-embedding alignment via generative reconstruction or contrastive distillation), and A-LLMRec / KAR [32] (attribute-aware MoE adapter); all operate on small CF datasets (ML-1M, Amazon-Books) and do not exploit structured metadata like genome tags. *Closest to our framework is RLMRec [23]*, which establishes alignment-loss content augmentation as a category; we differ in framing the choice—*regularizer* (aux-loss + ID; R1-gene/R1-plus), *replacer* (content-only; R2/R3), *injection* (ID+content; M4/M7), or *sequential* (interaction order; SASRec)—as a density-dependent decision, and in triangulating each class with two structurally-different instantiations. Recent methods map onto this taxonomy—e.g., topology-level augmentation (TAGCF [20]) and training-free descriptor-token mapping (FACE [28]) are injection/content-encoder variants (predicted density-robust); CF-knowledge injection (A-LLMRec [14]) and post-alignment ID/content fusion (Molar [18]) are regularizer-family strategies (predicted monotone in 1/density); and LLM-agent planning (RecMind [30]) defines an orthogonal agentic axis beyond our four; CF-embedding injection *into* the LLM (CoLLM [36]) and LLM-driven behaviour simulation that warms cold items (ColdLLM [12]) are further LLM-in-the-loop axes that carry serving-time LLM cost—but all are concurrent or evaluated on disjoint datasets under sequential/agentic/LLM-in-the-loop protocols rather than our full-ranking CF setup; we situate rather than benchmark them (a leaderboard would conflate paradigm with protocol), with our peer-reviewed RLMRec/KAR instantiations as established same-class anchors.

Text embeddings for RS. Sentence transformers [22] encode item descriptions, but naive title/genre encoding captures only surface lexicon; our LLM-as-synthesiser stage interposes compositional reasoning between metadata and embedding.

Multimodal augmentations of ML-20M. M3L-20M [26], MMTF-14K [5], KB4Rec [37], LLMRec [31], and ViLLA-MMBench [6] apply encoders to raw signals (ResNet, AST, CLIP) on partial coverage or smaller subsets (Table 1). LLM-MovieLens differs by interposing an *LLM reasoning stage* that jointly reads genome tags, plot, cast, and reviews, and synthesises embedding-optimised text capturing inter-attribute relationships. Advantages: 100% coverage of genome movies without imputation; structured multi-granularity output (profile + mood + themes); embedding-optimised design; \$14 reproduction cost. Complementary to—not competing with—perceptual augmentations.

3 The LLM-MovieLens Benchmark

3.1 Overview

The benchmark consists of a three-stage pipeline (Figure 1): (1) **LLM profile generation**, which synthesises embedding-optimised movie profiles from ML-20M genome tags and TMDb metadata; (2) **embedding generation**, which converts profiles into multiple feature types; and (3) **controlled evaluation**, which measures the impact of each feature type on recommendation quality.

3.2 Stage 1: LLM Profile Generation

Input data. For each of the 10,381 genome-covered movies in ML-20M, we assemble a prompt containing: (a) the top-30 genome tags ranked by relevance score, (b) TMDb metadata (plot overview, keywords, cast, crew, runtime, vote average), (c) MovieLens aggregate statistics (mean rating, rating count), and (d) user-generated tags. **Model, prompt, output.** Claude Haiku 4.5 (`claude-haiku-4-5-20251001`) at temperature 0.3 with a ∼1,327-token system prompt encoding five design principles for embedding quality (lead with thematic essence; exclude structured metadata that has separate feature columns; enforce consistent semantic flow; tight 80–120-word count; banned generic vocabulary). Output: a JSON record per movie with fields `movieId`, `title`, `profile` (80–120 words), `mood_vector` (10 axes on $[-1, 1]$), `key_themes` (3–5 strings), `word_count`. **Mood axes, QA, cost.** Mood axes are 10 bipolar perceptual dimensions in $[-1, 1]$ (e.g., `dark_light`, `serious_playful`, `realistic_fantastical`; full spec in App. A.2); per-genre signatures are interpretable (horror -0.56 on dark/light, comedy $+0.55$ on serious/playful). Every generation is JSON-schema- and constraint-validated with auto-retry; 22/10,381 required a regex pre-processor for numeric-title `movieId` corruption. Total cost: ∼ \$14 USD at ∼52 min runtime via Anthropic prompt caching.

3.2.1 *Human Evaluation.*

A 500-profile stratified sample (seed=42, genre-proportional) was rated by 3 graduate-student annotators, blinded to source LLM, on a 5-point Likert across five axes (Thematic Accuracy, Discriminativeness, Rule Compliance, Factual Consistency, Coherence & Fluency), preceded by a 30-profile pilot calibration; no post-hoc adjudication so inter-annotator agreement (κ) [2] reflects honest variance. Annotators were paid \$12/hour (∼\$216 total), above U.S. federal minimum and our institution's prevailing graduate rate; voluntary participation. Rating publicly available movie descriptions did not require IRB review (LLM-output evaluation, not human-subjects research).

Claude profiles outscore GPT-4o-mini on all five axes; the largest gaps are on Discriminativeness (+0.84 Likert) and Thematic Accuracy (+0.42). Discriminativeness is where Claude pulls furthest ahead: Claude profiles articulate movie-specific identity ($\geq$4 rate 71.9%), while GPT profiles often stop at generic genre summaries ($\geq$4 rate 30.9%). Factual Consistency is comparable because both models are prompted with identical TMDb summaries. Full guidelines, anchor scales, $\geq$ 4 rates per genre, ANOVA, and 100-pair mood pairwise validation are in App. A.3.

3.3 Stage 2: Embedding Generation

We convert Stage 1 outputs into six feature primitives. **LLM-derived:** *profile embeddings* (1024-d, encoded via `bge-large-en-v1.5` [33], L2-normalised, no PCA; used by M4/M7/M8/R1/R2); *mood vectors* (10-d, lifted from JSON; used by M5/M7/M8/M9/R2); *key themes* (528-d multi-hot over the 528 themes appearing in $\geq$ 10 movies; used by M6/M8/M9). **Baselines:** *genome PCA* (128-d at 80.0% explained variance; M2); *genome raw* (1,128-d, no PCA, controls for dimensionality; M2b); *BERT title* (1024-d, "Title | Genres" encoded with the *same* bge transformer; M3, isolating LLM reasoning from text encoding). **Composites:** M7 = profile $\oplus$ mood (1034-d), M8 = profile $\oplus$ mood $\oplus$ themes (1562-d), M9 = genome PCA $\oplus$ mood $\oplus$ themes (666-d). A shared side-feature MLP maps any input dimension to the model's 128-d embedding space.

3.4 The 10-Axis Mood Vector as a Controllable-Recommendation Primitive

Mood merits separate treatment because its value lies in the *new tasks the artifact enables*, not in marginal NDCG gains: adding mood to profile (M7 vs. M4) is statistically null on top-K (Q4: ΔNDCG@10 = +0.2%, n.s.). 10 named bipolar axes cannot beat 1024 dense profile dimensions where dense semantic similarity already saturates. We validate mood along three orthogonal axes.

Controllability (zero-shot). For each of 100 query items and 10 mood axes, we blend profile-cosine (0.7) with mood-proximity (0.3) after offsetting the query's mood vector by +0.5 along the target axis. Mean controllability score (centroid shift / requested offset) is 30.8% (range 18–51%, baseline $\approx$ 0); off-axis drift 0.17. Most controllable: `realistic_fantastical` (51.2%). Zero-shot: no training, no separate retrieval index.

Compactness (per-dim efficiency). M5 (mood-only, 10-d) recovers ∼64% of M2's medium-bucket Recall@1000 (0.063, genome-PCA-128d) at 1/13 the dimensionality, and ∼90% of M3's (0.044, BERT-on-title-1024d) at 1/100 the dimensionality—an 8.2× per-dim advantage over genome-PCA and 92× over BERT-on-title. On the cold bucket, mood loses to both (0.0007 vs. 0.0044/0.0020).

Cross-LLM stability (per-axis). Re-generating all 10,381 mood vectors with GPT-4o-mini [21] at the identical prompt and computing Pearson r per axis: mean r = 0.586 with sharp variation—four HIGH-stability axes ($r \geq 0.7$: `dark_light` 0.91, `serious_playful` 0.90, `realistic_fantastical` 0.73, `slow_fast` 0.71), five MEDIUM-stability, one LOW (`nostalgic_contemporary`,

Table 1: Comparison of content-augmented resources for MovieLens. Coverage is fraction of base-dataset items with features. LLM-MovieLens is the only resource providing LLM-synthesised semantic features with structured multi-granularity output (†M3L-20M coverage when all four modalities required; text-only 92%. ‡100% of genome-covered movies; 38% of all ML-20M).

Resource	Modalities	Items	Cov.	LLM
ML-20M Genome	Tag scores	10,381	38%	—
M3L-20M	Text/Img/Aud/Vid	19,009	71%†	—
PosterLens 20M [1]	Image	27,163	99.6%	—
MMTF-14K	Audio+Visual	13,623	50%	—
KB4Rec	Knowledge graph	25,982	95%	—
LLMRec (ML-10M)	Text+Image	ML-10M	—	✓
ViLLA-MMBench	Text/Aud/Vis	∼14K	50%	✓
LLM-MovieLens (ours)	**Text+Mood+Themes**	**10,381**	**100%‡**	✓

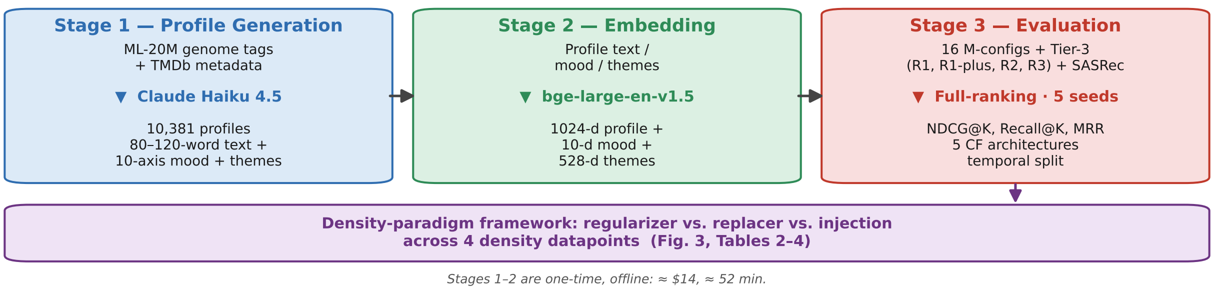

Figure 1: End-to-end pipeline. Stage 1 synthesises LLM profiles from genome tags + TMDb metadata (one-time, offline); Stage 2 encodes them with bge-large-en-v1.5 and derives mood/theme features; Stage 3 evaluates the 16 M-configs together with the Tier-3 LLM-for-RecSys methods (R1, R1-plus, R2, R3) and SASRec under full-ranking, 5 seeds, temporal split. Stage-3 outputs across four density datapoints feed the density-paradigm framework—the paper's central contribution—analysed in Fig. 3 and Tables 5 and 6. The ▼ marks the transforming model/method within each stage; horizontal arrows are inter-stage data flow.

Table 2: Human evaluation results on 500 stratified profiles (3 annotators, seed=42). κ_F: Fleiss' kappa; κ_W: weighted kappa; ≥ 4: fraction rated *good* or *excellent*. Annotators were blinded to source LLM.

	Claude Haiku 4.5				GPT-4o-mini			
Axis	Mean±Std	κ_F	κ_W	≥ 4	Mean±Std	κ_F	κ_W	≥ 4
Thematic Acc.	4.12±.74	.30	.40	80.9%	3.70±.76	.03	.03	61.2%
Discrim.	3.96±.83	.27	.40	71.9%	3.12±.81	.02	.05	30.9%
Rule Compl.	4.58±.57	.47	.52	97.2%	4.27±.68	.01	.01	87.4%
Factual	4.23±.68	.42	.50	86.1%	4.21±.69	.02	.02	85.7%
Coh. & Flu.	4.44±.60	.50	.54	94.3%	4.27±.69	.03	.02	87.9%

$r = -0.074$, anti-correlated). Deployment-safe subset: the four HIGH axes (or multi-LLM ensemble for the rest).

3.5 Stage 3: Evaluation Protocol

We restrict ML-20M to its 10,381 genome-covered movies (preserving 99% of ratings), convert to implicit feedback (≥ 3.5 stars), apply iterative 10-core filtering, and split by timestamp (train: pre-2014-01-01, 99.0%/11.5M; val: 2014-01-01 to 2014-07-01, 0.4%/49.7K; test: post-2014-07-01, 0.6%/67.5K), yielding 11,486,938 training interactions across 127,053 users and 9,906 items at 0.91% density. We perform *full ranking* (score all 9,906 items per test user, mask training items) and report NDCG@K, Recall@K for $K \in \{10, 20, 50\}$, and MRR; we avoid sampled metrics following [17]. Items are partitioned by training interaction count into *cold* (<10, 112 items), *medium* (10–49, 1,872 items), and *warm* (≥ 50, 7,922 items) for stratified analysis. All experiments use 5 random seeds (42, 123, 456, 789, 2026); we report mean ± std and paired t-tests. Cross-hardware variance contributes $\leq 25\%$ of total 5-seed spread (App. A.4); paired t-tests cancel per-seed hardware drift.

4 Experimental Setup

4.1 Baseline Models

We organise models into three tiers based on content-feature use, spanning a classical-to-contrastive CF spectrum so improvements cannot be attributed to a weak baseline.

Tier 1: Pure CF (no content). M0: BPR-MF [25]; M1: LightGCN [9] (3-layer, mean pooling); M1b, M1c, M1d:

Table 3: The paradigm taxonomy mapped to its instantiations and density predictions. Each content-paradigm class is triangulated by *two* structurally-different instantiations; sequential is an orthogonal axis (§6).

Class	Instantiations	Density prediction
Injection	M4, M7	density-robust (reference)
Regularizer	R1-gene, R1-plus	gains as density falls
Replacer	R2, R3 (no ID)	loses at every density; collapses sparse
Sequential	SASRec	orthogonal; steepest decay

SimGCL [35], XSimGCL [34], LightGCL [4]—three state-of-the-art graph-contrastive variants covering the current pure-CF frontier.

Tier 2: Content-augmented LightGCN-SF (primary ablations). The item embedding fed into graph propagation is $\mathbf{e}_i^{(0)} = \mathbf{e}_i^{\text{learned}} + \text{MLP}(\mathbf{f}_i)$, with $\mathbf{e}_i^{\text{learned}} \in \mathbb{R}^d$ the learnable item embedding, $\mathbf{f}_i$ the content feature vector, and a 2-layer MLP (feat_dim$\to d \to d$, ReLU). The 9 Tier-2 configs (M2–M9) are listed in Table 4; feature primitives and composites are defined in §3.3.

Tier 3: LLM-for-RecSys methods. The taxonomy (Table 3) is a contribution; each class is triangulated by two structurally-different instantiations. **R1** (RLMRec-gene) [23] and **R1-plus** (RLMRec-plus) [23] are the regularizer pair (generative reconstruction vs. contrastive InfoNCE alignment of LLM-content and CF embeddings; both on all four densities, App. A.9.3). **R2** (KAR-style MoE replacer, HEA-LightGCN, inspired by [32]; we adapt KAR's HEA from CTR to a BPR backbone, App. A.5) and **R3** (HypernetReplacer, $\mathbf{e}_i = \text{MLP}(\mathbf{c}_i)$, no ID, no gating) are the replacer pair, isolating the gating mechanism. **SASRec** [13] (faithful pmixer, upstream config) is the orthogonal sequential axis.

Protocol asymmetry, disclosed. R1 uses RLMRec's upstream per-dataset YAMLs. R2's hyperparameters come from a grid spanning KAR's published per-task ranges (a faithful-instantiation requirement, since R2 adapts KAR); R3, a paper-original minimal replacer with no upstream, uses a compact grid identical to R2's per-dataset retune grid, making R2-vs-R3 a controlled comparison. All three are tuned on ML-20M (their home dataset); per-dataset re-tuning is a robustness check applied only to the held-out density datapoints (ML-1M, Amazon-Books)—ML-20M is not re-tuned. R1's tuning advantage makes our framework claims conservative; R2's Amazon collapse cannot be explained by it (R1 is the gainer there). Full grids and retune analyses in App. A.8, App. A.9.1, App. A.9.2.

4.2 Research Questions

Seven questions organise the experiments; all are answered on NDCG@10 / Recall@10 / MRR.

Pre-specified ML-20M ablations (designed and frozen before any cross-density runs; rule out trivial confounds):

- **Q1** M4 vs. M1 / M1c — does LLM content beat pure CF and strong contrastive methods?
- **Q2** M4 vs. M2 PCA-128d / M2b raw 1128-d — does the gain survive a dimensionality control?
- **Q3** M4 vs. M3 at the same encoder — does LLM *reasoning* beat naive title encoding?
- **Q4** M7 vs. M4 — does mood add measurable signal on NDCG@10/Recall@10/MRR?

Cross-density synthesis (the density-paradigm framework is synthesised post-hoc from the four-datapoint chain—ML-20M 1,160, subsampled-ML-20M 225 same-domain, ML-1M 163, Amazon-Books 13 int/item—and presented as a falsifiable cross-density hypothesis, §6):

- **Q5** is the regularizer-vs-injection NDCG@10 gap monotone in 1/density, and do replacers (R2 + R3) fail to beat M7 at every density while collapsing when sparse?
- **Q6** does SASRec's dense-to-sparse decay rate differ from the content paradigms?

Robustness. Q7 encoder (bge vs. e5) and LLM-provider (Claude vs. GPT-4o-mini) sensitivity. Mood-primitive validation is non-Q1–Q7 (§3.4).

4.3 Hyperparameters

All models trained under our LightGCN-SF infrastructure (M0–M9, R2, R3) share: embedding dimension $d = 128$, Adam optimizer (lr $= 10^{-3}$), L2 weight decay 10^{-5}, batch size 32,768, 1 negative sample per positive (BPR). LightGCN variants use 3 propagation layers, no dropout. Contrastive models use $\lambda_{\text{CL}} = 0.2$, $\tau = 0.2$; SimGCL/XSimGCL use noise magnitude $\epsilon = 0.1$, LightGCL uses SVD rank $q = 5$. Training: up to 200 epochs, early stopping (patience 20, validation NDCG@10). Four configurations follow upstream protocols: R1, R1-plus, SASRec, and the KAR sanity check; details in App. A.8.

5 Results

5.1 Main Results

Table 4 (visualised in Fig. 2) reports mean ± std over 5 seeds (config labels defined in §3.3).

Q1: LLM content beats pure CF including strong contrastive CF.. M4 vs. M1 (LightGCN): +3.0% NDCG@10 ($p = 0.004$); vs. M1c (XSimGCL, strongest contrastive): +2.6% NDCG@10 ($p = 0.010$), MRR +3.5% ($p < 0.001$).

Q2: LLM beats genome regardless of dimensionality. M4 vs. M2 (PCA-128d): +2.5% NDCG@10 ($p = 0.001$); vs. M2b (raw 1128-d): +2.9% NDCG@10 ($p = 0.022$). M2b has more input dims than M4—rules out dimensionality.

Q3: LLM reasoning beats text-only encoding. M4 vs. M3 (BERT-on-title at the same encoder): +3.3% NDCG@10 ($p = 0.002$), +4.4% Recall@10 ($p = 0.006$)—isolates reasoning from encoding.

Q4: Mood is a near-null on top-10. M7 leads M4 on MRR by +1.0% ($p = 0.061$, marginal); NDCG@10 ($\Delta = +0.2\%$,

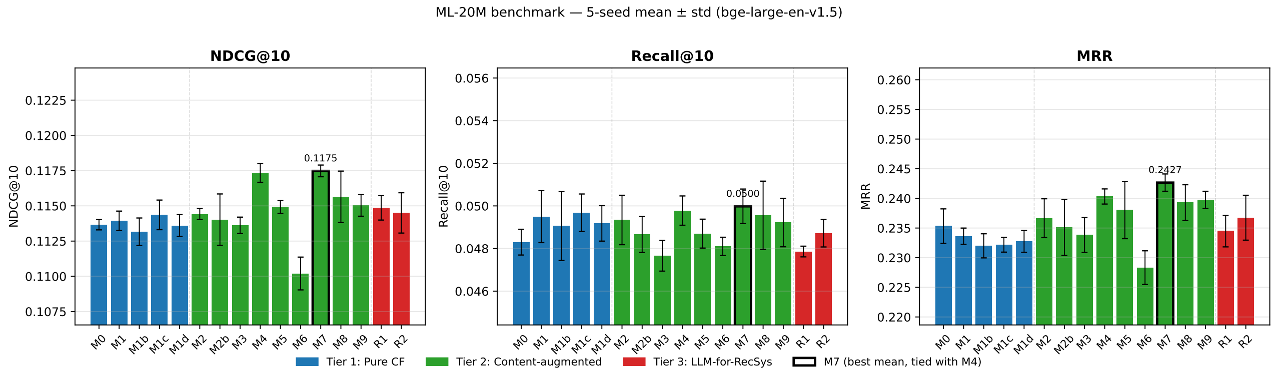

Figure 2: Main ML-20M benchmark (NDCG@10, Recall@10, MRR) for all 16 configurations, coloured by tier (blue: pure CF; green: content-augmented; red: LLM-for-RecSys). The LLM-augmented configs (M4, M5, M7, M9) form the top cluster; M7 has the highest mean on all three metrics but is statistically tied with M4 on NDCG@10/Recall@10 (only the marginal MRR lead, $p = 0.061$). Exact values and significance in Table 4 and Table 6.

Table 4: Main benchmark results on ML-20M test set. Best within each tier underlined. M4 and M7 are statistically tied on NDCG@10 and Recall@10 (paired t-test $p > 0.05$); M7 leads on MRR at $p = 0.061$ (marginal). We do not bold a single "overall best" on NDCG@10/Recall@10 because no configuration significantly dominates at $p < 0.05$.

Config	NDCG@10	Recall@10	MRR
Tier 1: Pure CF			
M0 BPR-MF	0.1137±.0004	0.0483±.0005	0.2357±.0022
M1 LightGCN	0.1139±.0007	0.0495±.0012	0.2336±.0014
M1b SimGCL	0.1132±.0010	0.0491±.0016	0.2320±.0020
M1c XSimGCL	0.1144±.0011	0.0497±.0009	0.2322±.0012
M1d LightGCL	0.1136±.0008	0.0492±.0008	0.2327±.0018
Tier 2: Content-Augmented LightGCN-SF			
M2 +Genome PCA(128)	0.1144±.0004	0.0493±.0012	0.2367±.0033
M2b +Genome Raw(1128)	0.1140±.0018	0.0487±.0008	0.2351±.0047
M3 +BERT Title(1024)	0.1136±.0006	0.0477±.0007	0.2338±.0029
M4 +LLM Profile(1024)	0.1173±.0007	0.0498±.0007	0.2403±.0013
M5 +LLM Mood(10)	0.1149±.0004	0.0487±.0007	0.2380±.0048
M6 +LLM Themes(528)	0.1102±.0012	0.0481±.0004	0.2283±.0028
M7 +Profile+Mood	0.1175±.0004	0.0500±.0008	**0.2427**±.0015
M8 +All LLM(1562)	0.1156±.0018	0.0496±.0016	0.2393±.0030
M9 +Genome+Mood+Themes	0.1150±.0008	0.0492±.0011	0.2397±.0015
Tier 3: LLM-for-RecSys			
R1 RLMRec-gene	0.1162±.0004	0.0495±.0007	0.2366±.0018
R2 KAR-MoE replacer	0.1145±.0014	0.0487±.0006	0.2367±.0038

n.s.) and Recall@10 are statistically tied. We do not designate M7 as overall best on top-10 ranking. The honest null: LLM-engineered profiles already capture most of the signal; mood may add a marginal lift at the very top of the ranking (MRR only).

Q5: Density determines paradigm choice (central finding). On ML-20M alone, M7 outperforms R1 by +1.1% NDCG@10 ($p = 0.007$) and R2 by +2.6% ($p = 0.013$). The full picture is Table 5 (Fig. 3): across the four densities the regularizer-vs-injection gap is *monotone in* $1/$*density for both instantiations* (R1-gene, R1-plus; every off-ML-20M datapoint $p < 0.001$;

Table 5: Consolidated class-level outcomes across density. NDCG@10 Δ% vs. the injection reference M7, with the *absolute* 5-seed-mean NDCG@10 in parentheses, at four per-item densities (int/item), for both structurally-different instantiations of each content-paradigm class (the M4/M7 row gives the injection reference's absolute NDCG@10; $\Delta \equiv 0$). Regularizers gain monotonically with sparsity; replacers lose at every density; injection is the density-robust reference. All non-tie entries are significant at $p < 0.05$ except R2 on ML-20M (n.s.); full p-values in Table 6 and Apps. A.9.3, A.9.2. Sequential SASRec (orthogonal axis) in Table 7. †same-domain density control.

Method (class)	1,160	225†	163	13
M4/M7 (injection, ref)	.1175	.1076	.1661	.0563
R1-gene (regularizer)	−1.1 (.1162)	+3.7 (.1116)	+17.0 (.1943)	+29.5 (.0729)
R1-plus (regularizer)	−5.6 (.1110)	+10.9 (.1193)	+16.3 (.1932)	+26.4 (.0711)
R2 (replacer)	−2.6 (.1145)	−5.1 (.1021)	tie (.1680)	−22.0 (.0439)
R3 (replacer)	−6.5 (.1099)	−3.9 (.1034)	tie (.1669)	−11.7 (.0497)

App. A.9.3), while *both replacers (R2, R3) underperform M7 at every density*—losing at three of four and tying only at mid density (ML-1M; App. A.9.2). The Q5 replacer sub-question thus answers *no, not uniformly*: replacers never *beat* M7 at any density. Injection (M4/M7) is density-robust (the R2−R3 sign-reversal mechanism is analysed in §6).

Q6: Sequential is an orthogonal axis with the steepest density decay. SASRec ties M7 on dense ML-20M (+0.2%). Within each dataset, SASRec vs. M1 decays monotonically across the three full-scale densities: +3.3% → −11.7% → −48.5% (ML-20M → ML-1M → Amazon; paired-t p = 0.031/<0.001/<0.001, $n = 5$). The absolute cross-dataset ratio is 6.8× from ML-20M to Amazon vs. M7's 2.1×—the steepest of any paradigm tested.

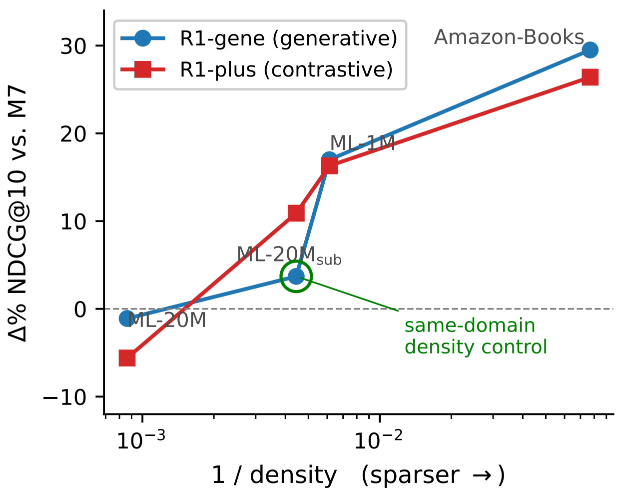

Figure 3: Visual companion to the all-class digest (Table 5), zooming in on the *regularizer* class: NDCG@10 gap (Δ% vs. the M7 injection config) across four per-item densities for both regularizer instantiations (R1-gene, R1-plus). Both are monotone in 1/density—losing at dense scale, winning by +26–30% at the sparse extreme. The circled point is the same-domain density control (subsampled-ML-20M: identical items/content as full ML-20M), isolating density from the density–domain confound. Above 0 the regularizer wins; below, injection wins. Per-datapoint p-values in App. A.9.3.

Q7: Encoder-robust; provider-effect detected, an order of magnitude below the content-effect. e5-large-v2 preserves the M7≥M4≥M3≈M8 ranking with ≤ 1.3% NDCG@10 spread. GPT-4o-mini retrain preserves M4≈M7 and M4-over-baseline rankings on every metric; a significant cross-provider effect on M4 NDCG@10 is detected: −0.73% ($p = 0.026$, $n = 5$, Claude > GPT every seed)—an order of magnitude below the M4-over-baseline gap. Claude profiles are markedly preferred by human raters on Discriminativeness (+0.84 Likert, Table 2), yet this large quality gap translates into only a −0.73% downstream NDCG@10 effect—evidence that injection-paradigm downstream metrics are robust to substantial content-quality variance.

5.2 Cold-Start Analysis

A matched-architecture ablation (M1 ID-only vs. M4 LLM profile vs. M7 profile+mood) on the 0.91%-density graph isolates the LLM content channel. On items with <10 training interactions, full-ranking pushes cold NDCG@10/Recall@10 to ≈ 0 for every method (protocol artefact), so we report Recall@1000 per bucket. **(i) Cold items become reachable**—R@1000 rises from 0 (M1) to 0.0033±.0005 (M4) and 0.0040±.0020 (M7), bootstrap CIs above zero. **(ii) Strongest signal is medium-bucket R@1000** (+28% M4, +45% M7 over M1's 0.0307). **(iii) Warm gain matches Table 4**: +3% NDCG@10 for M4/M7. Amazon-Books reveals an additional regime: aux-loss methods (R1) break the cold-bucket NDCG@10 ceiling (R1 cold NDCG@10 = 0.0057, 3.6× M4/M7) by reshaping CF embeddings rather than only adding features (App. A.9). Fig. 4 shows the per-bucket rescue: the medium bucket is where LLM content lifts most.

6 Analysis

The cross-density experiments synthesise into a falsifiable framework predicting which LLM-for-RecSys paradigm wins at which density regime (Tables 3, 5): aux-loss *regularizers* gain monotonically with sparsity; capacity *replacers* (both instantiations, all four datapoints) never beat M7 and lose at every density, collapsing when sparse; additive *injection* (M4/M7) is density-robust. Mechanisms, visualisations, and robustness checks in App. A.

Triangulation per content class, with within-class crossover. Both content classes are corroborated across *all four* density datapoints (incl. the same-domain control) by a second structurally-different instantiation—R1-plus (contrastive distillation, slope matched to R1-gene; App. A.9.3) for regularizers, R3 (gating-free; App. A.9.2) for replacers. The R2−R3 sign reverses and localises to the densest point alone (R2>R3 only on ML-20M)—MoE-with-CF-gating is a bidirectional CF-signal amplifier, not a one-way loss-amplifier. *Falsifiers:* a future regularizer that does not gain at sparse-per-user density, a replacer that does not lose, or a dataset where R1's gain over M7 is non-monotone refutes the corresponding prediction.

Orthogonal axis: sequential. SASRec ties M7 on dense ML-20M and falls 6.8× to sparse Amazon-Books—the steepest dense-to-sparse endpoint drop of any paradigm we test (App. A.9). **Why LLM synthesis beats raw genome:** M4 beats M2b at higher dim (+2.9%, $p = 0.022$, Table 6); LLM vs. genome top-10 nearest-neighbour retrieval overlap is 1.0/10 on average—profile geometry differs structurally from genome geometry.

Robustness. Encoder (e5-large-v2 within ≤ 1.3% NDCG@10, App. A.6); cross-LLM (−0.73% residual on M4 NDCG@10 under GPT-4o-mini, App. A.7); cross-platform (R2 MPS vs. CUDA ±0.002 AUC, App. A.5); per-dataset retuning (R2 still loses on Amazon under a 4-cell grid, App. A.9.1); variance decomposition (App. A.4).

7 Discussion and Limitations

LLM features are most valuable for sparse-history items, metadata-rich domains, and offline precomputation; \$14 one-time cost scales linearly (∼\$1,400/1M items), with prompt-only adaptation to e-commerce, music, and papers. LLM-MovieLens is *complementary* to perceptual datasets like

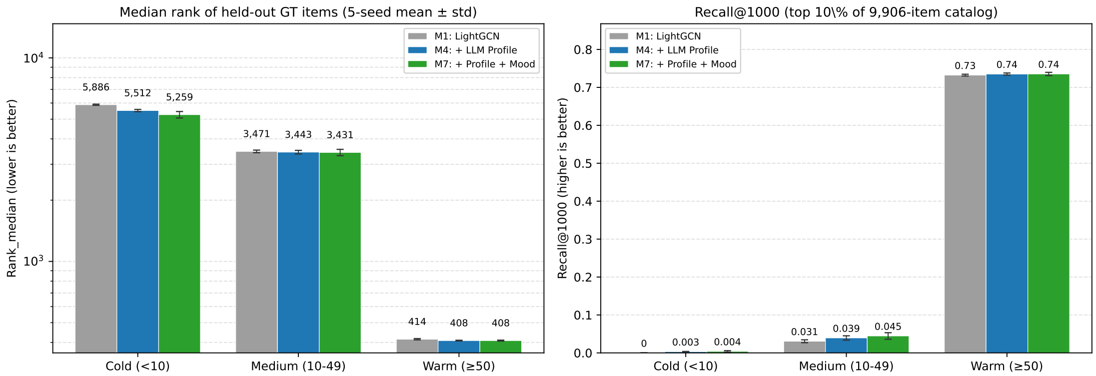

Figure 4: Cold-start analysis on ML-20M (5-seed mean ± std; 1σ error bars), matched-architecture M1 vs. M4 vs. M7. Left: median rank of held-out items by bucket (lower is better). Right: Recall@1000 by bucket. The medium-bucket rise from 0.031 (M1) to 0.039 (M4, $+28\%$) and 0.045 (M7, $+45\%$) is the strongest absolute rescue; the cold bucket lifts from exactly 0 to ≈ 0.004. Source values: §5.2.

M3L-20M [26]. Five downstream directions—cold-start enrichment, zero-shot controllable retrieval via the mood primitive, cross-domain transfer, density-adaptive paradigm switching, and reproducibility benchmarking—are detailed in the public artifact repository.

Limitations. ML-20M genome covers only 10,381/27,278 movies (out-of-genome items rely on TMDb alone); implicit-feedback thresholding ($\geq$ 3.5) discards explicit negative signal; the framework's empirical support is bounded to four density datapoints (ML-20M, subsampled-ML-20M same-domain control, ML-1M, Amazon) across two domains (movies, books) plus the orthogonal SASRec axis; broader coverage (e.g., Yelp, Steam, cross-lingual catalogues) and additional paradigm instantiations would further strengthen generality. *On aggregate-statistic inputs:* prompts include two MovieLens aggregate fields (mean rating, rating count) computed over the full history, but the generation system prompt forbids the output profile from referencing ratings, counts, popularity, or awards (principle P2, released), so only thematic synthesis enters the embedded text; the headline density comparison is moreover leakage-invariant by construction, since regularizer/injection/replacer configs consume identical features. *On tuning asymmetry:* R1's per-dataset YAMLs vs. R2/R3's leaner grids advantage regularizers, making the chain claim conservative—under matched grids the dense-endpoint gap could only narrow, since replacers already lose at every density. *Threshold:* ≥ 3.5 follows RLMRec's protocol for comparability; sensitivity is deliberate follow-up. *Deferred validations:* four planned follow-ups on the released testbed—a no-aggregate-statistics prompt ablation (leakage-invariance predicts a negligible effect); a content-dominant replacer with a small residual-ID pathway (separating replacer collapse from ID absence per se); stronger sequential baselines (DuoRec, BERT4Rec) beyond SASRec; and a mood-primitive controllability user study.

8 Broader Impact

LLM-MovieLens removes the LLM-API serving-time barrier for content-augmented recommendation: a one-time $14 generation step replaces per-query LLM inference, eliminating the recurring inference-time energy and latency cost that a deployed LLM-in-the-loop recommender would incur. Profiles may inherit TMDb / MovieLens biases (the full generation log is released for audit), and downstream recommendation systems built atop the artifact remain subject to standard recsys harms (filter bubbles, popularity bias, demographic skew in genre coverage); downstream users should treat the artifact as inheriting these risks and audit accordingly. Licensing: ML-20M is research-only per its original terms; TMDb metadata follows TMDb TOS; LLM profiles and pre-computed embeddings are released under CC BY 4.0; code under MIT. A full Gebru et al. [7]-style datasheet, Croissant 1.0 metadata, and a SHA-256 manifest covering all released artefacts ship with the artifact.

9 Conclusion

LLM-MovieLens releases an extensible testbed (10,381 ML-20M profiles + parallel ML-1M / Amazon-Books packs, multi-encoder embeddings, 16 model configurations) and uses it to establish a density-paradigm framework for LLM-for-RecSys, triangulated by two structurally-different instantiations per content-paradigm class (regularizer: R1-gene + R1-plus; replacer: R2 + R3) so claims are class-level, not method-specific. On ML-20M, LLM profiles beat genome, same-encoder BERT-title, and pure CF (incl. contrastive) at $p < 0.05$. Across four density datapoints the regularizer-vs-injection gap is monotone in 1/density for *both* regularizer instantiations—R1-gene $-1.1\% \to +29.5\%$ and R1-plus $-5.6\% \to +26.4\%$ vs. M7, including the subsampled-ML-20M same-domain density control—while replacers never beat M7 and collapse when sparse, injection is density-robust, and SASRec's 6.8$\times$ drop is the steepest. **Implication: paradigm choice should adapt to per-item density rather than remain fixed**

across benchmarks. Three directions follow: a density-adaptive paradigm-switcher trained on the released testbed; cross-domain replication beyond movies/books (Yelp, Steam, Goodreads) via the released prompt template; and cross-domain controllable retrieval through the mood primitive—each a falsification opportunity for these class-level claims on future densities and instantiations.

A Extended Experimental Results

A.1 Paired t-tests for Headline Comparisons

Table 6: Paired t-tests (5 seeds, bge-large-en-v1.5). Significance: $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$. The five within-ML-20M LLM-content comparisons (Q1–Q3; M4 vs. each baseline) all remain significant on NDCG@10 under Holm–Bonferroni correction ($\alpha =$ 0.05).

Comparison	NDCG@10		Recall@10		MRR	
	Δ	p	Δ	p	Δ	p
M4 vs. M1 (LightGCN)	+.0034	**	+.0003	ns	+.0067	**
M4 vs. M1c (XSimGCL)	+.0030	*	+.0001	ns	+.0081	***
M4 vs. M2 (PCA-128)	+.0028	**	+.0005	ns	+.0037	ns
M4 vs. M2b (Raw-1128)	+.0033	*	+.0011	**	+.0052	ns
M4 vs. M3 (BERT)	+.0037	**	+.0021	**	+.0065	*
M7 vs. M4 (+mood)	+.0001	ns	+.0002	ns	+.0023	ns
M4 vs. R1	+.0012	*	+.0003	ns	+.0037	*
M4 vs. R2	+.0028	*	+.0011	ns	+.0036	ns
M7 vs. R1	+.0013	**	+.0005	ns	+.0061	*
M7 vs. R2	+.0030	*	+.0013	*	+.0059	*

A.2 Mood Vector Specification

The 10 bipolar axes—dark/light, serious/playful, slow/fast, cerebral/visceral, realistic/fantastical, intimate/epic, conventional/experimental, emotional/detached, nostalgic/contemporary, predictable/subversive—are each a continuous value in $[-1, 1]$; full named-pole definitions are released in the public artifact repository. Per-genre signatures are interpretable: horror clusters dark/serious (-0.56 on dark/light), comedies in light/playful ($+0.55$ on serious/playful), documentaries grounded (-0.58 on realistic/fantastical), action visceral and epic ($+0.53$, $+0.25$).

A.3 Human Evaluation Details

A 30-profile pilot calibration preceded the main study; no post-hoc adjudication, so κ reflects honest variance. Genre-stratified ANOVA: no significant inter-genre difference on Coherence & Fluency ($p = 0.21$); a marginal effect on Discriminativeness driven by the Documentary genre ($p = 0.04$, $\eta^2 = 0.06$). The 100-pair mood pairwise validation reaches 80.1% definite-direction agreement. Full 5-point Likert anchor definitions, annotation guidelines, and per-genre breakdowns are released in the public artifact repository.

A.4 Variance Decomposition (Seed vs. Hardware)

Headline configs were re-run on Apple-Silicon M2, NVIDIA RTX 3090/4090, and Colab T4/L4. Cross-platform delta on test NDCG@10 (seed=42, M7) is ≤ 0.0008; total 5-seed std is ± 0.0004–$.0018$. Hardware drift contributes $\leq 25\%$ of total 5-seed spread. Paired t-tests cancel per-seed hardware drift, so p-values in Table 6 are robust to which seed ran on which GPU.

A.5 KAR Sanity Check on Native CTR Domain

To verify R2 (HEA-LightGCN) implementation correctness independently of our adaptations, we ran the upstream KAR HEA on its native ML-1M CTR config on both Apple-Silicon (MPS) and CUDA T4. Test AUC agrees within ± 0.002 across platforms (0.798 MPS vs. 0.799 CUDA). This sanity-checks platform parity, not the BPR adaptation; the BPR-adapted R2 is a representative replacer-class instantiation, not a published-KAR reproduction.

A.6 Encoder Sensitivity

Re-running M3/M4/M7/M8 with `intfloat/e5-large-v2` [29] in place of `bge-large-en-v1.5`: NDCG@10 of M7 changes from $0.1175 \pm .0004$ to $0.1162 \pm .0006$, a 1.1% relative drop; M4/M3/M8 shift by $\leq 1.3\%$ NDCG@10. The ordering M7$\geq$M4$\geq$M3$\approx$M8 is preserved on every seed. We claim encoder-robustness within the bge/e5 cost tier tested. The check targets M3/M4/M7/M8 (encoder-dependent: feature input is a sentence-transformer encoding of LLM text); encoder-independent configurations (M1 / M2 / M5 / M6 / M9 / R2 / R3) need no re-run, and R1 / R1-plus consume the same M4-feature so their cross-encoder behavior is bounded by M4's.

A.7 LLM-Provider Sensitivity (Cross-LLM Downstream)

We regenerated all 10,381 profiles with GPT-4o-mini at the identical prompt and retrained M4/M7 with 5 seeds. **Ranking preserved:** M4 still beats genome-PCA (+1.6% NDCG@10), BERT-on-title (+2.4%), and LightGCN (+2.1%). **Residual on M4 NDCG@10:** -0.73% ($p = 0.026$, paired t, $n = 5$, Claude $>$ GPT every seed)—an order of magnitude below the M4-over-baseline gap. **Interpretation:** the +0.84 Likert Discriminativeness gap (Table 2) translates to only -0.73% downstream NDCG@10—injection-paradigm downstream metrics dampen content-quality variance through the GCN's CF-side ID embeddings. **What mediates the gap:** over all 10,381 items, Claude profiles carry 35.2% greater cross-item mood-vector spread (mean per-axis std 0.482 vs. 0.357) and are longer/more lexically diverse (115 vs. 102 words; TTR 0.830 vs. 0.762)—the wider mood separation is what raters reward as discriminativeness, yet maps to only the -0.73% downstream effect. The check targets M4/M7 because the cross-LLM perturbation is upstream of the downstream method: R1 / R1-plus / R2 / R3 consume the *same* generated content features as M4/M7, so the cross-LLM effect propagates equally through their shared input. A single M4/M7 cross-LLM run therefore bounds the effect for the full Tier-3 / Tier-3+ stack.

A.8 Tier-3 Hyperparameter Selection

R1 inherits RLMRec's upstream per-dataset YAMLs; on its home dataset ML-20M we ran a 54-point native selection grid over `layer_num`, `mask_ratio`, `recon_weight`, and `re_temperature` ($2\times3\times3\times3$; `reg_weight` held fixed at the upstream default 10^{-7}, not tuned), selecting by best validation Recall@20—RLMRec's native upstream selection/early-stopping criterion (patience 5)—over 5 seeds. On ML-20M the grid runs at `embedding_size`=128 (capacity-matched to the shared d=128 backbone, not RLMRec's upstream 32); the other three datapoints keep RLMRec-gene's upstream-native 32-d and R1-plus uses 32-d throughout—an asymmetry that only makes the sparse-datapoint R1 gains more conservative. R2 uses an 18-point ML-20M grid (winner inherited unchanged to ML-1M/Amazon, matching the M-config protocol); R3 uses a 4-cell per-dataset grid (App. A.9.2). All grid axes, per-cell scores, winners, and per-seed test metrics are released under a unified per-method tree at `code/benchmark/hparams/r{1,2,3}/`: each directory carries a single `grid_selection.json` (the authoritative manifest) plus flat `<dataset>_winner_seed<N>.json` files and a `README.md`. R1's larger tuning budget makes our framework claims conservative; R2's Amazon collapse is not a tuning artefact—it still loses after per-dataset retuning (App. A.9.1).

A.9 Cross-Domain Validation on Amazon-Books-2018

We replicate the Tier-2 / Tier-3 pipeline on Amazon-Books-2018 [19] (Reviews split, 10-core filter), yielding 52,643 users, 91,599 items, 1,183,610 training interactions (13 int/item; 0.025% density). At this sparse extreme, regularizers gain dramatically while replacers collapse. Table 5 shows the 4-datapoint chain; per-dataset numbers below.

A.9.1 R2 Retune Robustness (R2 still loses). On Amazon-Books-2018, R2 with ML-20M-tuned defaults: −28.8% vs. M7. Per-dataset 4-cell grid retune (learning rate $\{10^{-3}, 3\times10^{-4}\}$ $\times$ weight decay $\{10^{-4}, 10^{-5}\}$) selects (lr $= 10^{-3}$, wd $= 10^{-4}$) and improves to −22.0% vs. M7 (still negative, $p < 0.001$). On the subsampled-ML-20M same-domain control (225 int/item) the same retune gives R2 −5.1% vs. M7 ($p < 0.01$, $n = 5$, 5/5 seeds same-sign). The replacer-class loss is not a tuning artefact.

A.9.2 R3 Triangulation (Hypernetwork Replacer). R3 ($\mathbf{e}_i = \mathrm{MLP}(\mathbf{c}_i)$, no ID embedding, no MoE/gating) on per-dataset 4-cell grid winners across all four densities: ML-20M −6.5% ($p = 0.021$), subsampled-ML-20M same-domain control −3.9% ($p < 0.001$), ML-1M ties, Amazon −11.7%. Both replacers thus lose to M7 at the same-domain control (R2 −5.1%, R3 −3.9%), completing replacer-class triangulation across all four datapoints. The R2−R3 sign reverses and localises to the densest point alone: R2>R3 by +4.0% only on ML-20M (CF-side MoE helps when CF signal is very strong); R3≥R2 at every lower density (subsampled-ML-20M, ML-1M, Amazon +10.3%). MoE-with-CF-gating is a bidirectional CF-signal amplifier, not a one-way loss-amplifier.

A.9.3 R1-plus Triangulation (Contrastive Regularizer). R1-plus (RLMRec-plus contrastive distillation) on all four density datapoints; per-dataset upstream-canonical hparams; ML-20M capacity-matched to d=128 mirroring R1-gene; depth-controlled Amazon re-run. NDCG@10 vs. M7 (all $n = 5$, 5/5 seeds same-sign): ML-20M −5.6% (0.1110 vs. 0.1175, $p = 6\times10^{-5}$; capacity-match narrows the d=32 −7.4% loss by 1.8 pp); subsampled-ML-20M same-domain control +10.9% (0.1193 vs. 0.1076, $p < 0.001$); ML-1M +16.3% ($p < 0.001$, within seed noise of R1-gene's +17.0%); Amazon +26.4% (vs. R1-gene's +29.5%). Both regularizer instantiations are therefore monotone in 1/density across the *full four-datapoint chain*: R1-gene −1.1% → +3.7% → +17.0% → +29.5%, R1-plus −5.6%→+10.9%→+16.3%→+26.4% (≈31–32 pp swing)—corroborated by a structurally-different second instantiation (generative reconstruction vs. contrastive InfoNCE) at every datapoint, eliminating the single-instantiation concern at the same-domain control.

A.9.4 SASRec Cross-Density. SASRec [13] is an orthogonal axis to the three content paradigms: it consumes interaction order, not item content. We reproduce SASRec (faithful pmixer [13]) on all four density datapoints to test whether the sequential paradigm follows the same density trajectory as content paradigms. **Result:** on dense ML-20M SASRec ties M7 (+0.2%, n.s.), but the dense-to-sparse drop is the steepest of any paradigm we test — 6.8× from ML-20M (0.1177) to Amazon-Books (0.0172), and −48.5% vs. M1 on Amazon. Subsampled-ML-20M (−2.2% vs. M1) shows the sequential axis is already brittle once per-item interactions drop below ∼225, well before crossing into a different domain. Sequential modelling is therefore not a free substitute for content augmentation in sparse regimes.

Table 7: SASRec (faithful pmixer) NDCG@10 across four densities, 5-seed mean ± std. Δ vs. M1 is within-dataset.

Dataset	NDCG@10	Δ vs. M1	vs. best content
ML-20M	0.1177±.0023	+3.3%	+0.2% vs. M7
Subsampled ML-20M	0.1048±.0060	−2.2% (n.s.)	−6.1% vs. R1
ML-1M	0.1469±.0039	−11.7%	−24.4% vs. R1
Amazon-Books	0.0172±.0007	−48.5%	−76.4% vs. R1

GenAI Usage Disclosure

Per CIKM 2026 policy, we provide a full disclosure of generative-AI use across data, code, analysis, and writing stages.

LLMs as objects of study (released artifacts). Two LLMs are evaluated and their outputs released as research artifacts:

- **Claude Haiku 4.5** (Anthropic, `claude-haiku-4-5-20251001`, temperature 0.3) was used to generate the 10,381 ML-20M profiles and parallel Amazon-Books-2018 (9,289) and ML-1M subset (2,788) profiles. Each profile is an 80–120-word semantic text plus a 10-axis mood vector and 3–5 themes. The system prompt is ∼1,327 tokens and encodes five design principles for embedding quality. Total generation cost: ∼\$14 USD via Anthropic prompt caching; ∼52 min wall-clock for the ML-20M pack.
- **GPT-4o-mini** (OpenAI, `gpt-4o-mini-2024-07-18`, temperature 0.3, identical prompt and constraints) was used to regenerate all 10,381 ML-20M profiles for cross-LLM sensitivity studies (App. A.7). Total cost: ∼\$12 USD.

LLMs as tools. No LLM was used in the paper text, experiment design, result interpretation, statistical analysis, or figure/table production—all prose, equations, and analytic claims are human-authored. A code-assistant LLM (Claude Code) helped with routine data-loading boilerplate, LaTeX template adaptation, and BibTeX formatting; all such code was reviewed and tested by the authors.

Quality assurance. LLM outputs (Claude Haiku 4.5 and GPT-4o-mini profiles) are JSON-schema- and constraint-validated with automatic retry on failure; 22/10,381 Claude profiles required a regex pre-processor to correct a numeric-title `movieId` corruption. Full generation logs (prompts, responses, retry traces, validation errors) are released with the artifact for audit.

Bias considerations. LLM-generated profiles may inherit biases present in their training data, including TMDb and MovieLens. We have not specifically audited these biases; downstream users should treat the artifact as inheriting standard recommendation-system bias risks (popularity bias, filter bubbles, demographic skew in genre coverage).

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